

THOMAS G. KIELY

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ACADEMIC POSITIONS

Sept. 2024 – Present	POSTDOCTORAL SCHOLAR Kavli Institute for Theoretical Physics
Supervisors	Cenke Xu, Matthew P. A. Fisher & Leon Balents

EDUCATION

Aug. 2018 – May 2024	PH.D. IN PHYSICS
Aug. 2018 – Mar. 2021	M.S. IN PHYSICS Cornell University
Thesis Advisor	<i>Phase Transitions and Transport Properties in Ultracold Atom Quantum Simulators</i> Erich Mueller
Aug. 2014 – May 2018	B.S. IN PHYSICS, B.A. IN ITALIAN, MINOR IN PHILOSOPHY Georgetown University
	<i>Summa Cum Laude</i>

PUBLICATIONS

Preprints:

- *Checkerboard Bose Hubbard Ladders using Transmon Arrays*
Pranjal Praneel, **TGK**, Andre G. Petukhov, Erich J. Mueller
[arXiv:2605.07906](#)
- *Observation of Strong-to-Weak Spontaneous Symmetry Breaking in a Dephased Fermi Gas*
Si Wang*, **TGK***, Dorothee Tell, Johannes Obermeyer, Marnix Barendregt, Petar Bojovic, Philipp M. Preiss, Abhijat Sarma, Titus Franz, Matthew P. A. Fisher, Cenke Xu, and Immanuel Bloch
[arXiv:2604.16137](#)
- *Strong-to-Weak Symmetry Breaking in Open Quantum Systems: From Discrete Particles to Continuum Hydrodynamics*
Jacob Hauser*, Kaixiang Su*, Hyunsoo Ha*, Jerome Lloyd*, **TGK**, Romain Vasseur, Sarang Gopalakrishnan, Cenke Xu, Matthew P. A. Fisher
[arXiv:2602.16045](#)
- *Observation of disorder-induced superfluidity*
Google Quantum AI and collaborators
[arXiv:2512.21416](#)

Published:

13. *Exact kinetic propagators for coherent state complex Langevin simulations*
TGK*, Ethan C. McGarrigle* and Glenn H. Fredrickson
[Phys. Rev. A **113**, 023310 \(2026\)](#)
[arXiv:2508.11057](#)
12. *Phase transition of topological index driven by dephasing*
TGK and Cenke Xu
[Phys. Rev. B **112**, L201112 \(2025\)](#)
[arXiv:2506.04102](#)
11. *Effective Conformal Field Theory generated from Pure and Dephased Chern insulator*
Abhijat Sarma, Yimu Bao, Nayan Myerson-Jain, **TGK** and Cenke Xu
[Phys. Rev. B **112**, 085130 \(2025\)](#)
[arXiv:2503.11788](#)
10. *Spin Liquid and Superconductivity emerging from Steady States and Measurements*
Kaixiang Su, Abhijat Sarma, Marcus Bintz, **TGK**, Yimu Bao, Matthew P. A. Fisher and Cenke Xu
[Phys. Rev. Lett. **135**, 050403 \(2025\)](#)
[arXiv:2408.07125](#)

9. *Vacancy-assisted superfluid drag*
TGK, Chao Zhang and Erich J. Mueller
[Phys. Rev. A **111**, 053302 \(2025\) \[Editor's Suggestion\]](#)
[arXiv:2502.00542](#)
8. *Continuous Wigner-Mott transition at $\nu = 1/5$*
TGK and Debanjan Chowdhury
[Phys. Rev. B **110**, L241112 \(2024\) \[Editor's Suggestion\]](#)
[arXiv:2305.13355](#)
7. *Multi-site gates for state preparation in quantum simulation of the Bose Hubbard Model*
Pranjal Praneel, **TGK**, Erich J. Mueller and Andre G. Petukhov
[Phys. Rev. A **110**, 062615 \(2024\)](#)
[arXiv:2410.00973](#)
6. *High-temperature transport in the one-dimensional mass-imbalanced Fermi-Hubbard model*
TGK and Erich J. Mueller
[Phys. Rev. A **109**, 063318 \(2024\)](#)
[arXiv:2404.08076](#)
5. *Role of conservation laws in the density matrix renormalization group*
TGK and Erich J. Mueller
[Phys. Rev. B **106**, 235126 \(2022\)](#)
[arXiv:2207.03465](#)
4. *Strong Increase in Ultrasound Attenuation Below T_c in Sr_2RuO_4 : Possible Evidence for Domains*
Sayak Ghosh, **TGK**, Arkady Shekhter, F. Jerzembeck, N. Kikugawa, Dmitry A. Sokolov, A. P. Mackenzie and B.J. Ramshaw
[Phys. Rev. B **106**, 024520 \(2022\)](#)
[arXiv:2109.00041](#)
3. *Superfluidity in the one-dimensional Bose-Hubbard model*
TGK and Erich J. Mueller
[Phys. Rev. B **105**, 134502 \(2022\)](#)
[arXiv:2202.0066](#)
2. *Transport in the 2D Fermi-Hubbard Model: Lessons from Weak Coupling*
TGK and Erich J. Mueller.
[Phys. Rev. B **104**, 165143 \(2021\) \[Editor's Suggestion\]](#)
[arXiv:2106.04479](#)
1. *Relationship between the transverse-field Ising model and the XY model via the rotating-wave approximation*
TGK and J. K. Freericks
[Phys. Rev. A **97**, 023611 \(2018\)](#)
[arXiv:1711.04386](#)

HONORS AND AWARDS

May 2023	Douglas Fitchen Memorial Award to enable international travel to study, pursue research, or partake in Physics-related events
April 2023	DAMOP Travel Award to attend the APS DAMOP 2023 meeting in Spokane, Washington
May 2018	Undergraduate Research Award for depth and impact of written and oral presentation of undergraduate senior thesis
May 2018	Kidwell Medal for excellence in undergraduate Physics coursework
May 2018	Dante Award for excellence in undergraduate Italian coursework
May 2018	Phi Beta Kappa
April 2013	National Merit Finalist

TALKS & POSTERS

Invited talks:

2026: *Strong-to-Weak Spontaneous Symmetry Breaking in Dephased Fermi Gases*, **Cornell University**, Ithaca, NY ([link](#))

2024: *Bose-Hubbard Physics using Google Quantum AI Hardware*, **Google Quantum AI**, Santa Barbara, CA

2023: *High-Temperature Transport in Fermi-Hubbard systems*, **KITP**, Santa Barbara, CA

Selected Conference Talks:

2026: *Strong-to-weak spontaneous symmetry breaking in dephased quantum matter*, **APS Global Physics Summit**, Denver, CO

2023: *Transport in the 2D Fermi-Hubbard Model: Lessons from Weak Coupling*, **APS DAMOP Meeting**, Spokane, WA

2022: *Superfluidity in the 1D Bose-Hubbard model*, **APS March Meeting**, Chicago, IL

Selected Conference Posters:

2025: *Vacancy-assisted superfluid drag*, **Frontiers of Quantum Science with Atom/Molecule Tweezer Arrays**, Aspen, CO

2023: *Transport in the mass-imbalanced 1D Fermi-Hubbard model*, **APS DAMOP Meeting**, Spokane, WA

2022: *Transport in the 2D Fermi-Hubbard Model: Lessons from Weak Coupling*, **International Conference on Atomic Physics**, Toronto, ON

2021: *Superfluidity in the 1D Bose-Hubbard model*, **Boulder Summer School**, Boulder, CO

TEACHING EXPERIENCE

Spring 2023	PHYS-2214: PHYSICS III: OSCILLATIONS, WAVES, AND QUANTUM PHYSICS Lead two discussion sections and a lab section, created weekly quizzes, graded homework, proctored and graded exams
Fall 2023	PHYS-7653: STATISTICAL PHYSICS II Graded and wrote solutions to problem sets
Spring 2022	PHYS-2213: PHYSICS II: ELECTROMAGNETISM Lead three discussion sections and twice weekly office hours, graded homework, proctored and graded exams
Fall 2020	PHYS-2213: PHYSICS II: ELECTROMAGNETISM Course held entirely online; lead three discussion sections and twice weekly office hours, graded homework, proctored and graded exams
Spring 2020	PHYS-1102: GENERAL PHYSICS II Staffed a flipped, self-taught classroom for 15hrs per week, set up student-run labs, graded lab notebooks, proctored and graded exams. Taught extensively over Zoom due to COVID
Spring 2019	PHYS-2208: FUNDAMENTALS OF PHYSICS II Lead two discussion sections and a lab section, created weekly quizzes, graded homework, proctored and graded exams
Fall 2018	PHYS-1101: GENERAL PHYSICS I Staffed a flipped, self-taught classroom for 15hrs per week, set up student-run labs, graded lab notebooks, proctored and graded exams
Fall 2017	PHYS-251: INTERMEDIATE MECHANICS Held office hours, lead a weekly tutorial, graded problem sets, proctored exams
Fall 2015	PHYS-153: RELATIVITY AND QUANTUM PHYSICS Held office hours, graded problem sets

PRESS ON RESEARCH

Phys. Rev. B **104**, 165143 (2021) [*Editor's Suggestion*]

- ["Weak coupling shows flaw in strange metal model"](#) (Phys.org)

EXTRACURRICULARS AND OUTREACH

April 2020	CORNELL EXPANDING YOUR HORIZONS CONFERENCE Workshop Co-Leader: "Physics of Bubbles"
Jun. 2019 – May 2020	CORNELL GRADUATE PEER MENTOR
Aug. 2014 – May 2018	GEORGETOWN MEN'S VARSITY LIGHTWEIGHT ROWING
Aug. 2014 – May 2018	GEORGETOWN CIRCOLO ITALIANO Treasurer (2017-2018)
June 2015 – Dec. 2015	GEORGETOWN PHYSICS PEER ADVISOR

SKILLS

Programming Languages	Python, C++, Julia, Java, Wolfram Language (Mathematica)
Languages	English, Italian